

Chapter 4

Phi-Regret and Multicalibration

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In the previous chapters, the central algorithmic primitive was a fixed point: a regret minimizer over the set of deviations Φ proposes a deviation φ , and the learner plays a point $\mathbf{x}$ satisfying $\mathbf{x} = \varphi(\mathbf{x})$. This chapter introduces a different route to the same destination, due to Farina and Perdomo [FP26]. The route passes through forecasting.

At a high level, there are two reductions. First, multicalibration can be obtained from ordinary external regret minimization over a class of tests $\mathcal{H}$, provided that we can solve an *expected variational inequality* (EVI). Second, Φ -regret can be obtained from a multicalibrated forecaster by best responding to the forecasts. Taken together, these reductions give a path from external regret to Φ -regret that does not require optimizing over the geometry of valid endomorphisms, nor computing their fixed points.

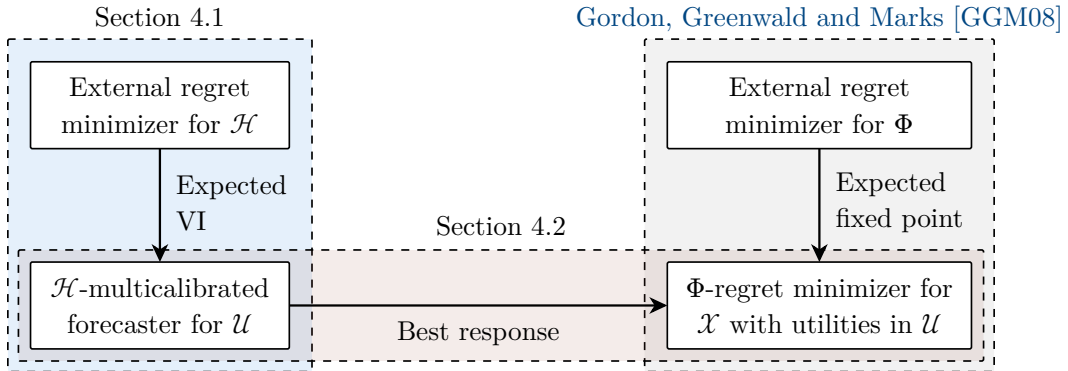


Figure 1: The two forecasting reductions and their relation to the Gordon-Greenwald-Marks framework.

4.1 A Gordon-Greenwald-Marks Theorem for Multicalibration

We start with online multicalibration. In every round t , Nature reveals a context $\mathbf{c}^{(t)} \in \mathcal{C}$. The learner outputs a distribution $D^{(t)}$ over forecasts $\mathbf{p}^{(t)} \in \mathcal{U} \subseteq \mathbb{R}^d$, and Nature then reveals

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the true utility vector $\mathbf{u}^{(t)} \in \mathcal{U}$. Given a test $h : \mathcal{C} \times \mathcal{U} \rightarrow \mathbb{R}^d$, the calibration error against h is

$$\text{MC-Err}_{T(h)} := \sum_{t=1}^T \mathbb{E}_{\mathbf{p}^{(t)} \sim D^{(t)}} [\langle h(\mathbf{c}^{(t)}, \mathbf{p}^{(t)}), \mathbf{u}^{(t)} - \mathbf{p}^{(t)} \rangle]. \quad (1)$$

The forecasts are $\mathcal{H}$ -multicalibrated if $\text{MC-Err}_{T(h)} = o(T)$ for every $h \in \mathcal{H}$. If $\mathcal{H}$ is symmetric, meaning that $h \in \mathcal{H}$ implies $-h \in \mathcal{H}$, this one-sided definition is equivalent to the usual absolute-value version. The one-sided form is more convenient here because it lines up exactly with regret inequalities.

The nonlinear primitive replacing fixed points is the following.

Definition 4.1 (Expected variational inequality). Let $S : \mathcal{U} \rightarrow \mathbb{R}^d$ be an operator and let $\varepsilon > 0$. An ε -solution to the expected variational inequality induced by S is a distribution D over $\mathcal{U}$ such that

$$\mathbb{E}_{\mathbf{p} \sim D} [\langle S(\mathbf{p}), \mathbf{u} - \mathbf{p} \rangle] \leq \varepsilon \quad \forall \mathbf{u} \in \mathcal{U}. \quad (2)$$

One way to parse (2) is as a randomized self-consistency condition. The distribution D may place mass on several forecasts; nevertheless, after averaging over that randomness, no possible utility vector $\mathbf{u} \in \mathcal{U}$ has positive correlation with the residual direction $S(\mathbf{p})$ by more than ε . This is exactly what will make the regret analysis telescope. Efficient algorithms for EVIs are known for general compact convex sets under mild oracle access assumptions; for our purposes, the important point is that EVIs are a standalone optimization primitive, just as fixed points were in the Gordon-Greenwald-Marks theorem from Chapter 1.

Now suppose that we have an external-regret minimizer $R_{\mathcal{H}}$ whose decision set is the class of tests $\mathcal{H}$. In each round, it chooses a test $h^{(t)} \in \mathcal{H}$; after the utility vector is revealed, it receives the linear utility

$$g^{(t)}(h) := \mathbb{E}_{\mathbf{p}^{(t)} \sim D^{(t)}} [\langle h(\mathbf{c}^{(t)}, \mathbf{p}^{(t)}), \mathbf{u}^{(t)} - \mathbf{p}^{(t)} \rangle]. \quad (3)$$

The reduction is as follows.

Algorithm: Multicalibration from external regret

Input: An external regret minimizer $R_{\mathcal{H}}$ over the test class $\mathcal{H}$

NextForecast($\mathbf{c}^{(t)}$):

 Set $h^{(t)} := R_{\mathcal{H}}.\text{NextStrategy}()$

 Let $D^{(t)}$ solve the EVI with $S^{(t)}(\mathbf{p}) := h^{(t)}(\mathbf{c}^{(t)}, \mathbf{p})$:

$$\mathbb{E}_{\mathbf{p}^{(t)} \sim D^{(t)}} [\langle h^{(t)}(\mathbf{c}^{(t)}, \mathbf{p}^{(t)}), \mathbf{u} - \mathbf{p}^{(t)} \rangle] \leq \varepsilon_t \quad \forall \mathbf{u} \in \mathcal{U} \quad (4)$$

return $D^{(t)}$

ObserveUtility($\mathbf{u}^{(t)}$):

 Feed the linear utility $g^{(t)}$ in (3) to $R_{\mathcal{H}}$

Theorem 4.2 ([FP26]). Let $\text{Reg}_{\mathcal{H}}^{(T)}(h)$ be the external regret of $R_{\mathcal{H}}$ with respect to the sequence of utilities $g^{(1)}, \dots, g^{(T)}$, and let $\text{EVI}_T := \sum_{t=1}^T \varepsilon_t$. Then the forecasts produced by the algorithm above satisfy

$$\text{MC-Err}_{T(h)} \leq \text{Reg}_{\mathcal{H}}^{(T)}(h) + \text{EVI}_T \quad \forall h \in \mathcal{H}. \quad (5)$$

Proof. The EVI condition is invoked at the realized utility vector $\mathbf{u}^{(t)}$, so

$$g^{(t)}(h^{(t)}) \leq \varepsilon_t. \quad (6)$$

Therefore, for any comparator test $h \in \mathcal{H}$,

$$\begin{aligned} \text{MC-Err}_{T(h)} &= \sum_{t=1}^T g^{(t)}(h) \\ &\leq \sum_{t=1}^T (g^{(t)}(h) - g^{(t)}(h^{(t)})) + \sum_{t=1}^T \varepsilon_t \\ &= \text{Reg}_{\mathcal{H}}^{(T)}(h) + \text{EVI}_T. \end{aligned} \quad (7)$$

Thus, sublinear regret over $\mathcal{H}$, together with $\sum_t \varepsilon_t = o(T)$, gives sublinear multicalibration error. $\square$

This proof is intentionally reminiscent of the GGM reduction. There, the fixed-point condition lets us replace the utility of the played strategy by the utility of the deviation selected by the regret minimizer. Here, the EVI condition lets us upper bound the calibration gain of the selected test $h^{(t)}$. The rest is just external regret.

The analogy is not merely cosmetic: EVIs are also necessary. Suppose we already had an efficient $\mathcal{H}$ -multicalibrated forecaster. Fix a context $\mathbf{c} \in \mathcal{C}$ and a test $h \in \mathcal{H}$, and consider the EVI operator $S(\mathbf{p}) = h(\mathbf{c}, \mathbf{p})$. To solve this EVI, run the forecaster for T rounds with the same context $\mathbf{c}^{(t)} = \mathbf{c}$. After it outputs $D^{(t)}$, choose Nature's response as

$$\mathbf{u}^{(t)} \in \arg \max_{\mathbf{u} \in \mathcal{U}} \left\langle \mathbf{u}, \mathbb{E}_{\mathbf{p} \sim D^{(t)}} [S(\mathbf{p})] \right\rangle. \quad (8)$$

For any fixed $\mathbf{u}^* \in \mathcal{U}$, the choice in (8) gives

$$\mathbb{E}_{\mathbf{p} \sim D^{(t)}} [\langle S(\mathbf{p}), \mathbf{u}^{(t)} - \mathbf{p} \rangle] \geq \mathbb{E}_{\mathbf{p} \sim D^{(t)}} [\langle S(\mathbf{p}), \mathbf{u}^* - \mathbf{p} \rangle]. \quad (9)$$

Averaging the distributions $D^{(1)}, \dots, D^{(T)}$ uniformly therefore produces a distribution D with

$$\mathbb{E}_{\mathbf{p} \sim D} [\langle S(\mathbf{p}), \mathbf{u}^* - \mathbf{p} \rangle] \leq \frac{\text{MC-Err}_{T(h)}}{T} \quad \forall \mathbf{u}^* \in \mathcal{U}. \quad (10)$$

As T grows, multicalibration drives the right-hand side to 0. So an online multicalibrated forecaster gives a black-box EVI solver. This is the precise sense in which EVIs play, for multicalibration, the role that fixed points play for Φ -regret.

4.2 From Multicalibration to Phi-Regret Minimization

We now turn the arrow around and use forecasting to make decisions. Let $\mathcal{X} \subseteq \mathbb{R}^d$ be a compact convex action set, and let $\mathcal{U} \subseteq \mathbb{R}^d$ be a compact convex set of possible utility vectors. A deviation class is a family $\Phi \subseteq \{\varphi : \mathcal{X} \rightarrow \mathcal{X}\}$. If the learner outputs distributions $\mu^{(t)}$ over $\mathcal{X}$, its Φ -regret against a deviation $\varphi \in \Phi$ is

$$\Phi\text{Reg}_{T(\varphi)} := \sum_{t=1}^T \mathbb{E}_{\mathbf{x}^{(t)} \sim \mu^{(t)}} [\langle \varphi(\mathbf{x}^{(t)}) - \mathbf{x}^{(t)}, \mathbf{u}^{(t)} \rangle]. \quad (11)$$

The reduction asks the forecaster to predict the next utility vector. Given a forecast $\mathbf{p} \in \mathcal{U}$, define the deterministic best response

$$\sigma(\mathbf{p}) \in \arg \max_{\mathbf{x} \in \mathcal{X}} \langle \mathbf{x}, \mathbf{p} \rangle, \quad (12)$$

with ties broken by a fixed rule. If the forecaster outputs a distribution $D^{(t)}$ over forecasts $\mathbf{p}^{(t)}$, the decision maker plays the pushforward distribution $\mu^{(t)}$ induced by $\mathbf{x}^{(t)} = \sigma(\mathbf{p}^{(t)})$.

What tests should the forecaster be calibrated against? For each deviation $\varphi \in \Phi$, define

$$h_{\varphi(\mathbf{p})} := \varphi(\sigma(\mathbf{p})) - \sigma(\mathbf{p}), \quad \mathcal{H}_{\Phi} := \{h_{\varphi} : \varphi \in \Phi\}. \quad (13)$$

This definition is the whole reduction. The test h_{φ} measures the direction in utility space in which the deviation φ would improve over the best response to the forecast.

Algorithm: Φ -regret from multicalibration

Input: A forecaster over $\mathcal{U}$ multicalibrated with respect to $\mathcal{H}_{\Phi}$

NextStrategy():

Query the forecaster and receive a distribution $D^{(t)}$ over forecasts $\mathbf{p}^{(t)} \in \mathcal{U}$
return the pushforward distribution $\mu^{(t)}$ of $\mathbf{x}^{(t)} = \sigma(\mathbf{p}^{(t)})$

ObserveUtility($\mathbf{u}^{(t)}$):

Feed the realized utility vector $\mathbf{u}^{(t)}$ to the forecaster

Theorem 4.3 ([FP26]). If the forecaster in the algorithm above has multicalibration error MC-Err_T with respect to $\mathcal{H}_{\Phi}$, then the decision maker has

$$\Phi\text{Reg}_{T(\varphi)} \leq \text{MC-Err}_{T(h_{\varphi})} \quad \forall \varphi \in \Phi. \quad (14)$$

In particular, sublinear $\mathcal{H}_{\Phi}$ -multicalibration implies sublinear Φ -regret.

Proof. Fix any $\varphi \in \Phi$. Since $\mu^{(t)}$ is the pushforward of $D^{(t)}$ under $\mathbf{p} \mapsto \sigma(\mathbf{p})$,

$$\begin{aligned}
\Phi \text{Reg}_{T(\varphi)} &= \sum_{t=1}^T \mathbb{E}_{\mathbf{p}^{(t)} \sim D^{(t)}} [\langle h_{\varphi(\mathbf{p}^{(t)})}, \mathbf{u}^{(t)} \rangle] \\
&= \sum_{t=1}^T \mathbb{E}_{\mathbf{p}^{(t)} \sim D^{(t)}} [\langle h_{\varphi(\mathbf{p}^{(t)})}, \mathbf{u}^{(t)} - \mathbf{p}^{(t)} \rangle] \\
&\quad + \sum_{t=1}^T \mathbb{E}_{\mathbf{p}^{(t)} \sim D^{(t)}} [\langle h_{\varphi(\mathbf{p}^{(t)})}, \mathbf{p}^{(t)} \rangle].
\end{aligned} \tag{15}$$

The first term is exactly $\text{MC-Err}_{T(h_\varphi)}$. The second term is nonpositive, because

$$\langle h_{\varphi(\mathbf{p})}, \mathbf{p} \rangle = \langle \varphi(\sigma(\mathbf{p})), \mathbf{p} \rangle - \langle \sigma(\mathbf{p}), \mathbf{p} \rangle \leq 0 \tag{16}$$

by the definition of $\sigma(\mathbf{p})$ as a maximizer over $\mathcal{X}$. This proves the claim. $\square$

The theorem is useful because the complexity of the required calibration class scales with the complexity of the deviation class. If Φ contains only constant deviations, then $\mathcal{H}_\Phi$ is a class of constant best-response gaps and we recover external regret. If Φ grows toward all maps $\mathcal{X} \rightarrow \mathcal{X}$, then $\mathcal{H}_\Phi$ becomes a strong calibration class and the result approaches the classical connection between calibration and swap regret.

The contextual version is identical. If contexts $\mathbf{c}^{(t)}$ are observed before play and deviations have the form $\varphi : \mathcal{C} \times \mathcal{X} \rightarrow \mathcal{X}$, then we define

$$h_{\varphi(\mathbf{c}, \mathbf{p})} := \varphi(\mathbf{c}, \sigma(\mathbf{p})) - \sigma(\mathbf{p}). \tag{17}$$

Multicalibration with respect to these tests gives contextual Φ -regret.

4.2.1 Putting the two reductions together

Combining Theorem 4.1 and Theorem 4.1 gives the promised route from external regret to Φ -regret. Set $\mathcal{H} = \mathcal{H}_\Phi$. Run the multicalibration algorithm as the forecaster inside the best-response algorithm. Then, for every $\varphi \in \Phi$,

$$\Phi \text{Reg}_{T(\varphi)} \leq \text{MC-Err}_{T(h_\varphi)} \leq \text{Reg}_{\mathcal{H}}^{(T)}(h_\varphi) + \text{EVI}_T. \tag{18}$$

Thus the burden of Φ -regret minimization shifts to two primitives:

1. no-regret learning over the induced test class $\mathcal{H}_\Phi$;
2. solving EVIs over the forecast domain $\mathcal{U}$.

The key distinction from the Gordon-Greenwald-Marks path is that we do *not* need to optimize over valid deviations directly. We only need a regret minimizer over tests that contain the maps h_φ . This extra flexibility is what makes the approach especially clean for large structured deviation classes.

4.2.2 Linear deviations

As an illustration, consider linear deviations $\varphi(\mathbf{x}) = \mathbf{M}\mathbf{x}$ over a convex compact set $\mathcal{X} \subseteq \mathbb{R}^d$. The classical GGM approach asks us to understand the geometry of all matrices $\mathbf{M}$ satisfying

$\mathbf{M} \mathcal{X} \subseteq \mathcal{X}$, and then compute fixed points of the matrices selected by the regret minimizer. That geometry can be difficult.

The multicalibration route permits a relaxation. From (13),

$$h_{\varphi(\mathbf{p})} = (\mathbf{M} - \mathbf{I})\sigma(\mathbf{p}). \quad (19)$$

If every valid linear endomorphism has spectral norm at most S , then each test above lies in a Frobenius ball of radius on the order of $\sqrt{d}(S+1)$. Instead of learning over valid endomorphisms, we can learn over the larger class

$$\mathcal{H} := \{\mathbf{p} \mapsto \mathbf{A}\sigma(\mathbf{p}) : \|\mathbf{A}\|_F \leq \rho\}, \quad (20)$$

where ρ is chosen large enough to contain all tests h_{φ} . This larger class is easy: external regret over a Euclidean ball is handled by projected gradient descent. Indeed, the utility sent to the matrix learner at time t is linear:

$$g^{(t)}(\mathbf{A}) = \mathbb{E}_{\mathbf{p}^{(t)} \sim D^{(t)}} [\langle \mathbf{A}\sigma(\mathbf{p}^{(t)}), \mathbf{u}^{(t)} - \mathbf{p}^{(t)} \rangle] = \left\langle \mathbf{A}, \mathbb{E}_{\mathbf{p}^{(t)} \sim D^{(t)}} [(\mathbf{u}^{(t)} - \mathbf{p}^{(t)})\sigma(\mathbf{p}^{(t)})^\top] \right\rangle_F. \quad (21)$$

Consequently, if $\|\mathbf{x}\|_2 \leq B$ for $\mathbf{x} \in \mathcal{X}$ and $\|\mathbf{u}\|_2 \leq L$ for $\mathbf{u} \in \mathcal{U}$, the standard projected-gradient bound gives $O(BLS\sqrt{dT})$ linear-swap regret, up to constants. Under the usual normalization in which $\mathcal{X}$ is in John's position, this improves the dimension dependence of the recent semi-separation-based route of [Daskalakis, Farina, Fishelson, Pipis and Schneider \[Das+25\]](#), while avoiding semi-separation altogether.

4.2.3 RKHS deviations

The same idea extends beyond finite-dimensional linear classes. Suppose deviations $\varphi : \mathcal{C} \times \mathcal{X} \rightarrow \mathcal{X}$ lie in a vector-valued reproducing kernel Hilbert space with matrix-valued kernel

$$\Gamma((\mathbf{c}, \mathbf{x}), (\mathbf{c}', \mathbf{x}')) \in \mathbb{R}^{d \times d}. \quad (22)$$

For the contextual reduction, the relevant tests are

$$h_{\varphi(\mathbf{c}, \mathbf{p})} = \varphi(\mathbf{c}, \sigma(\mathbf{p})) - \sigma(\mathbf{p}). \quad (23)$$

These tests also lie in an RKHS. Indeed, the first term is represented by composing the deviation kernel with the best-response map, while the second term is represented by the linear kernel on $\sigma(\mathbf{p})$. The resulting kernel is

$$\Gamma'((\mathbf{c}, \mathbf{p}), (\mathbf{c}', \mathbf{p}')) := \Gamma((\mathbf{c}, \sigma(\mathbf{p})), (\mathbf{c}', \sigma(\mathbf{p}'))) + \sigma(\mathbf{p})\sigma(\mathbf{p}')^\top. \quad (24)$$

So, if we have an online learner over the RKHS ball associated with Γ' , Theorem 4.1 supplies a multicalibrated forecaster, and Theorem 4.1 turns it into a no- Φ -regret algorithm. This recovers low-degree polynomial deviations as a special case and also covers infinite-dimensional kernels, such as Gaussian kernels, where the fixed-point route does not have an obvious finite-dimensional endomorphism geometry to exploit.

The moral is that forecasting separates the two hard-looking parts of Φ -regret. Best response converts forecasts to actions; multicalibration certifies that no deviation can systematically

exploit the forecast errors; and EVIs provide the self-consistency condition that makes the forecaster black-box reducible to ordinary external regret.

■ Bibliography for this chapter

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